S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR

(An Autonomous Institute, Affiliated to R.T.M. Nagpur University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

To create competent and creative professional in the field of Data Science to address the data-driven decision-making for the benefits of industry and society

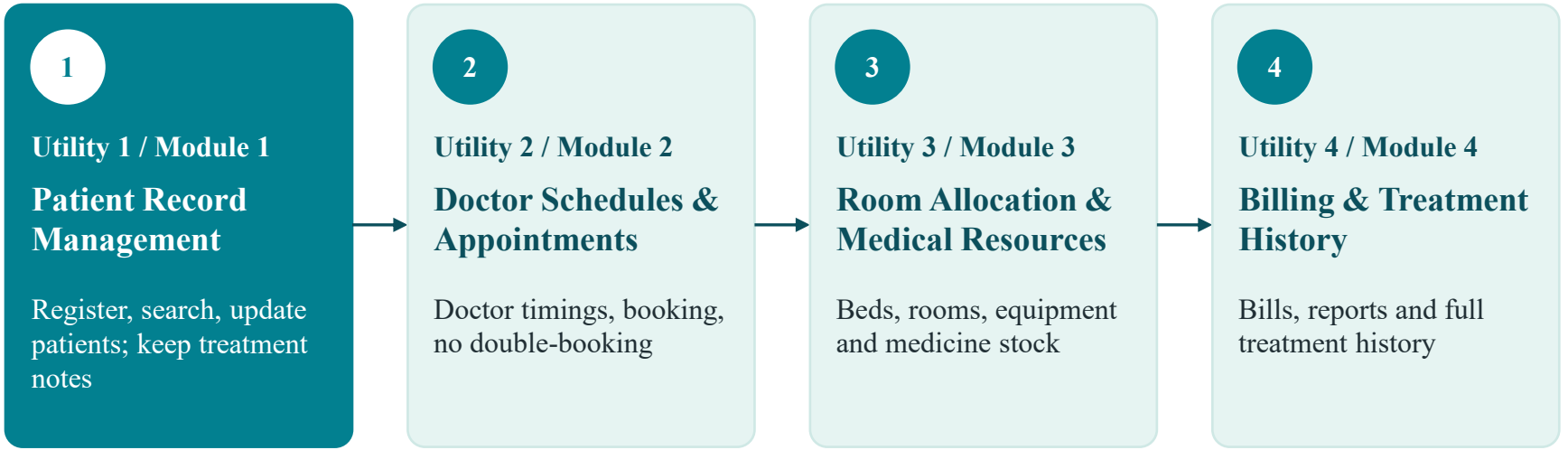
Aim: Develop a Python application to manage patient records, doctor schedules, appointments, room allocation, medical resources, billing, and treatment history while ensuring efficient hospital resource utilization.

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Project Roadmap: 4 Levels (Modules)

Overview

In simple words: one Python program that keeps a hospital's patients, doctors, rooms, medicines and bills in one place. I am building it step by step in 4 modules; each level uses the one before it.



This presentation covers Module 1 (Utility 1) - the base of the whole system.

Module 1 (Utility 1): Patient Record Management

Why first? Every other module (appointments, rooms, bills) needs a Patient ID, so patient records must be built first.

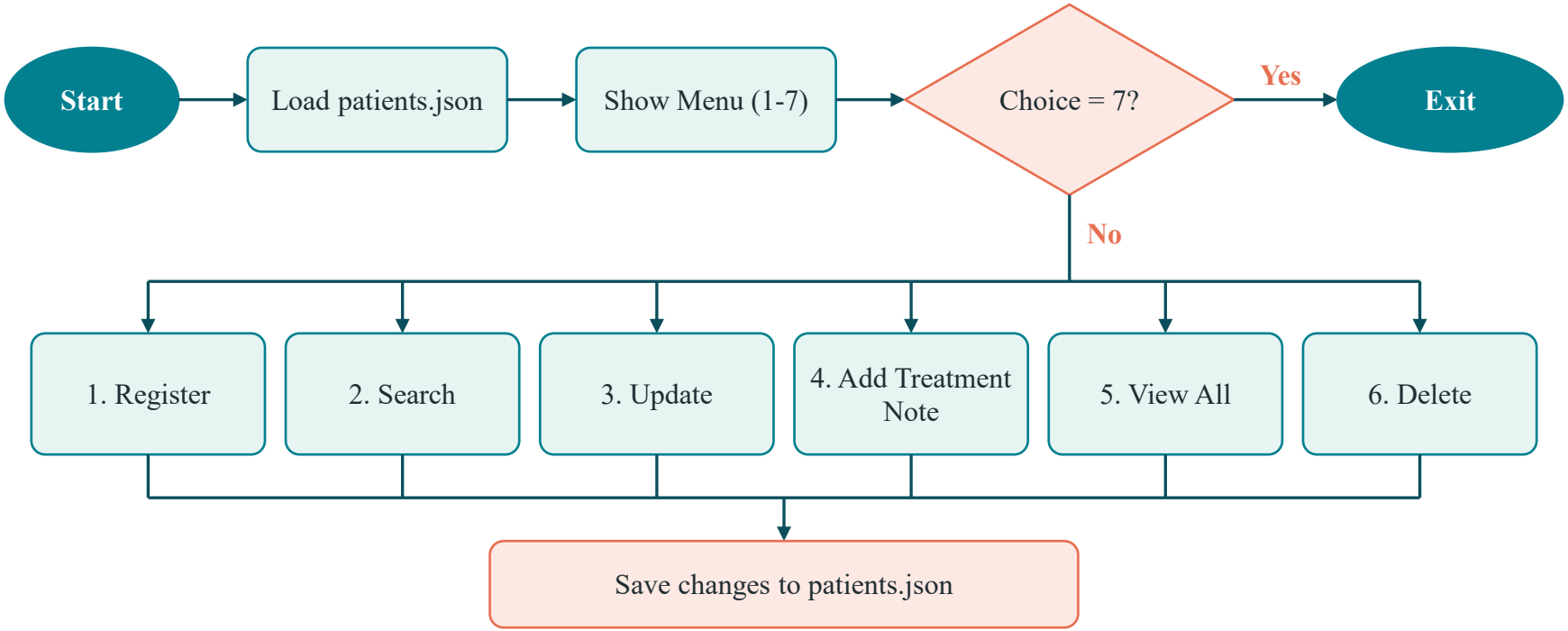
1 What Module 1 does

- Registers a new patient and gives a unique ID (P001, P002...)
- Searches a patient by ID or by name
- Updates details like name, age, phone
- Adds dated treatment notes (treatment history)
- Shows all records and deletes a record

2 What I did / am doing

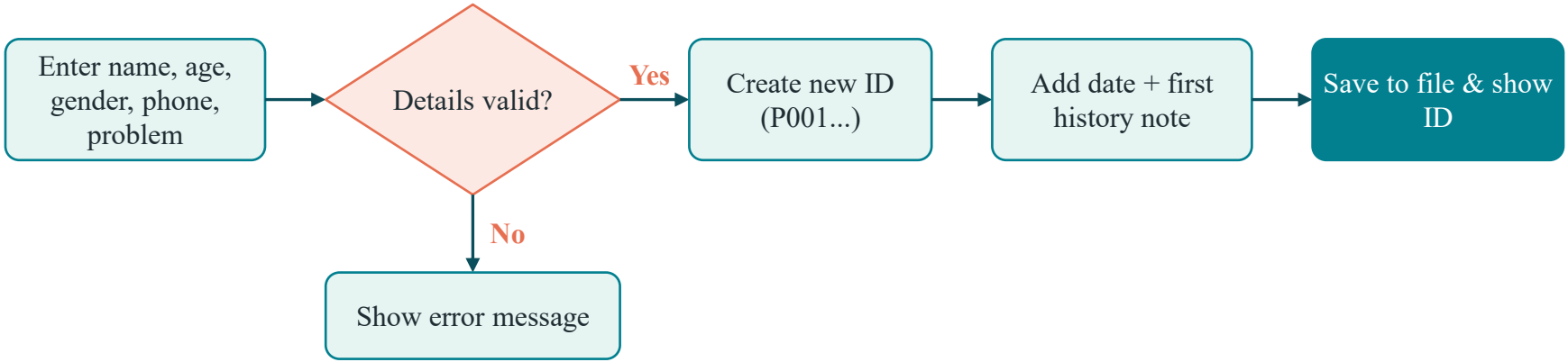
- Wrote the program in Python using functions and a dictionary
- Saved data in a patients.json file so it is not lost
- Added checks: valid age and 10-digit phone number
- Built a simple menu (options 1 to 7) for the user
- Tested it with sample patients - working

Flowchart 1: Overall Flow of Module 1



Then go back to the menu. This repeats until the user chooses 7 (Exit).

Flowchart 2 & Logic: How a Patient is Registered



Validation rules (checks)

- Name must not be empty
- Age must be between 1 and 119
- Phone must be exactly 10 digits

Search & update logic

- Search: match ID exactly, or part of the name
- Update: only name, age, phone, gender allowed
- Every treatment note is stored with its date

Requirements and Sample Output

What is required

- Python 3 (with IDLE / VS Code)
- Built-in modules only: json, os, datetime
- Input: patient details typed by the user
- Storage: patients.json file (acts as a small database)
- Output: patient list and history on the screen
- Files: module1_patient_records.py

Sample output

```
=== PATIENT RECORD MANAGEMENT ===
1.Register 2.Search 3.Update 4.Add Note
5.View All 6.Delete 7.Exit
Choice: 1
Registered! ID = P001

Choice: 2   Enter ID or name: ravi
P001 | Ravi Kumar | 34 | Male | 9876543210
  - 29-09-2026: Fever
  - 29-09-2026: Prescribed paracetamol
```

Conclusion and Next Step

- Module 1 (Utility 1) is done: the system can now store, find, update and remove patient records safely.
- Each patient has a unique ID that the next modules will use.
- Data is saved in a file, so nothing is lost when the program closes.

Coming next

Utility 2: Doctors & Appointments

Utility 3: Rooms & Resources

Utility 4: Billing & History

Thank you